RATING METHODOLOGY

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Analyst Contacts

Brian Oak +1.212.553.2946
MD-Corporate Finance
brian.oak@moodys.com

Matthew Moore +61.2.9270.8108
Associate Managing Director
matthew.moore@moodys.com

Karen Nickerson +1.212.553.4924
Associate Managing Director
karen.nickerson@moodys.com

Rating Methodology

Mining

This rating methodology replaces the *Mining* methodology published on 21 October 2021. While this methodology reflects many of the same core principles as the 2021 methodology, we have made changes to the scorecard to provide greater consistency of approach across our corporate sector methodologies. In the Leverage and Coverage factor, we have removed the Debt/Book Capitalization sub-factor and replaced the sub-factors EBIT/Interest Expense and (CFO-Dividends)/Debt with (EBITDA-Capex)/Interest Expense and RCF/Debt, respectively. We have changed some factor and sub-factor weights and scoring thresholds. We have also made limited editorial updates to some sections of the methodology, and we have changed the presentation of the scorecard.

Scope

This methodology applies to companies globally that are primarily engaged in the extraction or mining of ore, coal or other minerals, and the subsequent smelting and refining of minerals and ore. For example, an aluminum company that mines bauxite, refines it into alumina and then smelts into aluminum would be rated using this methodology. Some mining companies also have downstream activities, such as packaging and manufacturing.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

Companies that are primarily engaged in purchasing metal as a raw material for rolling and further value-added processing (rather than extracting metals from the ground as part of their own operations) and that derive revenue and earnings from a margin-on-metal construct are rated using our methodology for the steel industry.[1] In a margin-on-metal construct, a company converts a metal into a higher-margin product.

Sector overview

Mining companies are a diverse group of issuers. Scale and cost-competitiveness, diversity of commodity exposure and geographic breadth are differentiating qualities. These companies principally produce base metals, such as copper, nickel, aluminum and zinc; precious metals, such as gold and silver; thermal and metallurgical coal; and other extractive minerals, such as iron ore, uranium, cobalt and tin.

The mining industry is characterized by cyclicality and high earnings volatility. Supply and demand for specific commodities, global economic growth levels and expectations — particularly for China, which generally accounts for at least half of the global production and consumption of a number of base and precious metals — and industrial production levels drive cyclical demand and price volatility. Trading activity also has a meaningful impact on price volatility.

The mining industry is also capital-intensive. The depletion of ore bodies and changes in ore grades over the life of a mine require fairly steady reinvestment in mine development and expansion for companies to sustain and increase production.

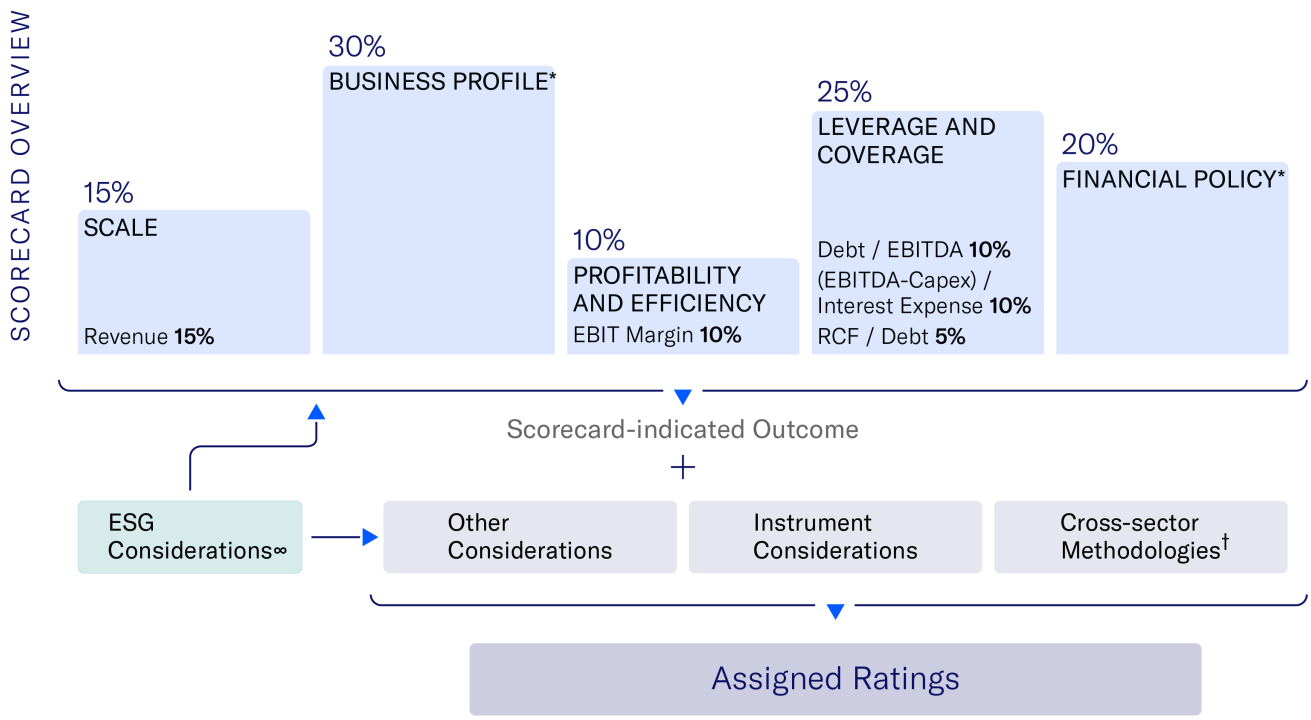
These challenges are common across the industry; however, earnings performance can vary widely among companies. Prices for base metals, precious metals, coal, iron ore and other metals move independently, reflecting different fundamental drivers. For example, gold prices are typically influenced by geopolitical risk, investment demand and sentiment toward inflation and other economic factors, whereas industrial production is generally a more significant driver for prices of base metals and other minerals and metals.

Rating approach

In this rating methodology, we explain our general approach to assessing credit risk of issuers in the mining industry globally, including the qualitative and quantitative factors that are likely to affect rating outcomes in this sector. We seek to incorporate all material credit considerations in ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

The following schematic illustrates our general framework for the analysis of mining companies, which includes the use of a scorecard. The scorecard-indicated outcome is not expected to match the actual rating for each company. For more information, see the “Other considerations” and “Limitations” sections.

Exhibit 1
Illustration of the mining methodology framework



* This factor has no sub-factors.
∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.
† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.
Source: Moody's Ratings

Mining scorecard

For general information about how we use the scorecard and for a discussion of scorecard mechanics, please see the "Using the scorecard to arrive at a scorecard-indicated outcome" section. The scorecard does not include or address every factor that a rating committee may consider in assigning ratings in this sector. Please see the "Other considerations" and "Limitations" sections.

Exhibit 2

Mining scorecard

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca	C
Factor: Scale (15%)										
Revenue (USD Billion)[1]	15%	≥ $100	$50 - $100	$25 - $50	$10 - $25	$4 - $10	$1.5 - $4	$1 - $1.5	$0.5 - $1	< $0.5
Factor: Business Profile (30%)										
Business Profile	30%	Expected volatility in results is almost non-existent, supported by a commanding market position, entrenched cost-effectiveness, technological leadership and a well-balanced global reach; and essentially no environmental, regulatory or geopolitical risk.	Very low expected volatility in results, supported by a deeply entrenched and leading market position that is highly defensible through cost-effectiveness and technological leadership and global exposure; and low environmental, regulatory and geopolitical risk.	Low expected volatility in results, supported by a very strong market position in its relevant market, demonstrated and sustainable competitive advantages, insulation from raw material cost fluctuations and solid diversity characteristics; and limited environmental, regulatory and geopolitical risk.	Moderate expected volatility in results, supported by a strong market position in its most important geographic or product markets; vertically integrated with advantageous cost profile; good diversity characteristics provide a buffer against sudden/ unexpected shifts in demand; modest but manageable environmental, regulatory and geopolitical risk.	Products are largely undifferentiated and the marketplace is highly competitive, exposing the company to periods of heightened volatility; established market position, favorable cost profile and fair diversity characteristics, including moderate operational concentration; moderate environmental, regulatory and geopolitical risk.	Products are undifferentiated, competition is intense and customers are price-sensitive, making results highly volatile; company does not have an advantageous cost profile or other mitigating competitive advantage; high operational concentration; or operations are primarily in countries or sectors with high environmental, regulatory or geopolitical risk.	Results are expected to be extremely volatile; company has a modest market presence, few competitive advantages, may have higher costs than competitors; high operational concentration (three mines); operations are primarily in countries or sectors with very high environmental, regulatory or geopolitical risk.	Near-term results are very difficult to predict and expected to be extremely volatile; no material competitive advantage; very high operational concentration (fewer than three mines); operations are concentrated in countries or sectors with a very high level of environmental, regulatory or geopolitical risk.	Near-term results are expected to be extremely negative with no expectation of recovery; or no material competitive advantage.

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca	C
Factor: Profitability and Efficiency (10%)										
EBIT Margin[2]	10%	≥ 60%	35% - 60%	25% - 35%	20% -25%	15% - 20%	10% - 15%	5% - 10%	0% - 5%	< 0%
Factor: Leverage and Coverage (25%)										
Debt / EBITDA[3]	10%	≤ 0.5x	0.5x - 1x	1x - 2x	2x - 3x	3x - 4x	4x - 6x	6x - 7.5x	7.5x - 9x	> 9x
(EBITDA-Capex) / Interest Expense[4]	10%	≥ 20x	16x - 20x	9x - 16x	4.5x - 9x	2.5x - 4.5x	1.25x - 2.5x	0.25x - 1.25x	0x - 0.25x	< 0x
RCF / Debt[5]	5%	≥ 100%	75% - 100%	50% - 75%	30% - 50%	20% - 30%	10% - 20%	5% - 10%	2.5% - 5%	< 2.5%
Factor: Financial Policy (20%)										
Financial Policy	20%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.	Expected to have financial policies (including risk and liquidity management) that create imminent risk of debt restructuring.

[1] For the linear scoring scale, the Aaa endpoint value is $150 billion. A value of $150 billion or better equates to a numeric score of 0.5. The C endpoint value is zero. A value of zero equates to a numeric score of 21.5.
[2] For the linear scoring scale, the Aaa endpoint value is 85%. A value of 85% or better equates to a numeric score of 0.5. The C endpoint value is (5)%. A value of (5)% or worse equates to a numeric score of 21.5.
[3] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The C endpoint value is 12x. A value of 12x or worse equates to a numeric score of 21.5, as does a negative Debt/EBITDA value.
[4] For the linear scoring scale, the Aaa endpoint value is 25x. A value of 25x or better equates to a numeric score of 0.5. The C endpoint value is (0.5)x. A value of (0.5)x or worse equates to a numeric score of 21.5.
[5] For the linear scoring scale, the Aaa endpoint value is 140%. A value of 140% or better equates to a numeric score of 0.5. The C endpoint value is 0%. A value of 0% or worse equates to a numeric score of 21.5.
Source: Moody's Ratings

Discussion of the scorecard factors

In this section, we explain our general approach for scoring each scorecard factor or sub-factor, and we describe why they are meaningful as credit indicators.

Factor: Scale (15% weight)

Why it matters

Scale is an important indicator of the overall depth of a company's business and its success in attracting a variety of customers, as well as its resilience to shocks, such as sudden shifts in demand or rapid cost increases. Companies with larger scale generally have greater flexibility to manage their businesses under different price and demand scenarios, which is important in an industry that is cyclical. A large revenue base can also lead to economies of scale that lower overall costs for raw materials and to efficiencies for corporate administration of the business, in areas such as finance, legal, tax and accounting. Larger companies also tend to generate higher cash flow for capital reinvestment and debt reduction and have greater access to the capital markets, which can reduce the cost of capital.

How we assess it for the scorecard

Scoring for this factor is based on one sub-factor: Revenue.

Revenue

Scale is measured (or estimated in the case of forward-looking expectations) using total reported revenue in billions of US dollars.

Factor: Business Profile (30% weight)

Why it matters

The business profile of a mining company is important because it greatly influences its ability to generate sustainable earnings and operating cash flows. These are important considerations in an industry characterized by cyclicality and high price volatility. The business profile also provides an indication of the levers available to a company's management to adjust mining operations when product prices are low or are declining.

Core aspects of a mining company's business profile include the types of commodities it sells and markets it sells into, its competitive position (primarily driven by the company's cost-effectiveness), the locations of its operations, and market demand. Susceptibility to environmental, regulatory and political risk is another important aspect of a company's business profile.

Pricing trends for different metals can vary, so a company concentrated in a single metal is generally more vulnerable to price volatility than one that sells multiple metals. The combination of significant top-line volatility and a limited ability to pass on costs to customers make the quality of a mining company's ore bodies and its cost efficiency essential in order to maintain its competitive position and to better withstand downward price movements.

The location and number of a company's mines is also important because having multiple, discrete assets in different regions typically lessens the impact of strikes, equipment failures, operational event risks, or other events that could curtail production. Geographic diversity also helps mining companies mitigate the impact of trade barriers and tariffs on exports.

Environmental requirements, political concerns and industry regulations are important considerations because they can affect costs and demand. For example, environmental requirements could constrain the production of a domestic mining company to the benefit of a foreign producer. The development of mines in countries lacking a strong rule of law and sufficient infrastructure can result in increased costs and production delays. Also, governments in some regions are more likely to require a greater proportion of local ownership in domestic mines (including a share of the earnings from these resources), which can reduce a mining company's profitability and cash flow and can cause operational disruptions while disputes are resolved. Regulations concerning labor relations, land use, native title, pollution, water scarcity and water quality can all contribute to greater uncertainty and higher costs.

How we assess it for the scorecard

Scoring for this factor is based on our expectations for earnings and cash flow volatility, which is influenced by the diversity of a company's commodity exposure and operations, its competitive position and its geographic diversity.

In assessing a company's competitive position, which is driven primarily by its mining costs and ability to control other costs, we typically consider the grade, or quality, of its deposits, mineral recovery rates, raw material procurement policies, inventory

management and relationships with suppliers. We also may consider any technological advantages that contribute to effective cost-control.

We assess geographic diversity by considering the number and location of a company's mines. Companies with three or fewer mines generally score lower for this factor than companies with a larger number of mines, especially if operations are concentrated in one country or in a limited number of countries. We typically also consider the diversity of the types of mines and the commodities produced.

We consider the environmental and regulatory environment in a given country in both our assessment of geographic concentration and our overall scoring for this factor. We typically assess risks that emanate from regulation, government actions and major events in any jurisdiction that could negatively affect the commercial activities of a mining company, including geopolitical risks that could impede trade or disrupt commodity markets. Such risks include tightening environmental regulations, policies to reduce excess capacity, tariffs or other restrictions on imports or exports, and royalty or tax agreements or carried interest requirements.

Generally, we do not expect a given company's business profile to exactly match the attributes listed for a given scoring category. We typically assign the factor score based on the alpha category for which the issuer has the greatest number of characteristics. However, there may be cases in which one characteristic is sufficiently important to a particular issuer that it has a large influence on the factor score. For example, reliance on three or fewer mines is likely to limit the score for this factor to the B or Caa category, because the operating risk in these cases is typically very high.

Factor: Profitability and Efficiency (10% weight)

Why it matters

Profits matter because they are needed to generate sustainable cash flow and maintain a competitive position, which includes making investments in operations. The ratio of earnings before interest and taxes to revenue (EBIT Margin) provides an indication of the overall efficiency of a company's cost structure, which is an important consideration in a capital-intensive industry.

How we assess it for the scorecard

Scoring for this factor is based on one sub-factor: EBIT Margin.[2]

EBIT Margin

The numerator is earnings before interest and taxes (EBIT), and the denominator is revenue.

Factor: Leverage and Coverage (25% weight)

Why it matters

Leverage and coverage measures provide important indications of a company's financial capacity and long-term viability. These metrics are important indicator of a company's ability to sustain its competitive position, access capital to invest in its business and service debt through different phases of the economic cycle.

This factor comprises three sub-factors:

Debt / EBITDA

The ratio of total debt to earnings before interest, taxes, depreciation and amortization (Debt/EBITDA) is an indicator of debt serviceability and financial leverage. The ratio is commonly used in this sector as a proxy for comparative financial strength.

(EBITDA-Capex) / Interest Expense

The ratio of EBITDA minus capital expenditures to interest expense ((EBITDA-Capex)/Interest Expense) indicates a company's ability to meet its interest obligations and invest in fixed assets.

RCF / Debt

The ratio of retained cash flow to debt (RCF/Debt) is an indicator of a company's cash generation (before working capital movements and capital expenditures, and after dividend payments) relative to its total debt.

How we assess it for the scorecard
Scoring for this factor is based on three sub-factors: Debt/EBITDA; (EBITDA-Capex)/Interest Expense; and RCF/Debt.[3]

Debt / EBITDA

The numerator is total debt, and the denominator is EBITDA.

(EBITDA-Capex) / Interest Expense

The numerator is EBITDA minus capital expenditures, and the denominator is interest expense.

RCF / Debt

The numerator is retained cash flow, and the denominator is total debt.

Factor: Financial Policy (20% weight)

Why it matters
Financial policy encompasses management and board tolerance for financial risk and commitment to a strong credit profile. It is an important rating determinant, because it directly affects debt levels, credit quality, the future direction for the company and the risk of adverse changes in financing and capital structure.

Financial risk tolerance serves as a guidepost to investment and capital allocation. An expectation that management will be committed to sustaining an improved credit profile is often necessary to support an upgrade. For example, we may not upgrade the ratings of a company that has built flexibility within its rating category if we believe the company will use that flexibility to fund a strategic acquisition, cash distribution to shareholders, spinoff or other leveraging transaction. Conversely, a company's credit rating may be better able to withstand a moderate leveraging event if management places a high priority on returning credit metrics to pre-transaction levels and has consistently demonstrated the commitment to do so through prior actions. Liquidity management is an important aspect of overall risk management and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which is discussed in the "Other considerations" section.

How we assess it for the scorecard
We assess the issuer's desired capital structure or targeted credit profile, its history of prior actions, including its track record of risk and liquidity management, and its adherence to its commitments. Attention is paid to management's operating performance and use of cash flow through different phases of economic and industry cycles. Also of interest is the way in which management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges or regulatory pressures. Considerations include a company's public commitments in this area, its track record of adhering to commitments and our views on the ability of the company to achieve its targets.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential negative impact of M&A or other types of balance-sheet-transforming events. Management's appetite for M&A activity is assessed, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. Frequency and materiality of acquisitions and previous financing choices are evaluated. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider negative repercussions caused by shareholders' willingness to sell the company.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. A track record of favoring shareholder returns at the expense of debtholders is likely to be viewed negatively in scoring this factor.

Other considerations

Ratings may reflect consideration of additional factors that are not in the scorecard, usually because the factor's credit importance varies widely among the issuers in the sector or because the factor may be important only under certain circumstances or for a subset of issuers. Such factors can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends, can also affect ratings.

The following are examples of additional considerations that may be reflected in our ratings and that may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and sub-factors in the scorecard.

Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given issuer is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, please see our methodology that describes our general principles for assessing ESG risks.[4]

The mining industry faces numerous environmental risks across a broad array of its operations. Regulations are becoming increasingly complex and stringent, and can include environmental stewardship and reclamation requirements, air and water quality and emissions standards, as well as native title issues, which can restrict land use and have other impacts. Changes in regulations can increase costs, extend project timelines and add uncertainty to the level of reserves that can be economically mined. All operations, including changes to existing operations, require numerous permits and licenses. Additional environmental considerations include litigation, reputational risks, product replacement risks and competitor strategies.

Environmental and regulatory risks that we may assess include the following:

- Reclamation and remediation requirements, which are typically spread out over many years. Various jurisdictions may require letters of credit or surety bonds, or may set financial performance metrics for self-bonding.
- Air pollution requirements, relating to emissions from mining equipment, smelting operations, or in the end-use of the commodity, for example.
- Soil, water and waste management requirements (for building tailings dams or disposing of hazardous or toxic substances, for example).
- Limits on carbon dioxide emissions. The aluminum smelting industry, for example, is highly energy-intensive, and the aluminum production process also has fairly high greenhouse gas emissions. The ability to reduce energy intensity can lower costs as well as carbon dioxide emissions, which can improve a company's competitive position.
- Increasing competition with local communities and agriculture for limited water resources. Smaller, less-diversified companies are the most vulnerable to competition for water because they lack the resources available to larger, lower- cost producers. Additionally, the use of water by mining companies is increasingly subject to objections from local communities.
- Natural and human-made disasters such as tailings dam and pit wall collapses, seismic activity and structural risks in deep underground mining activity.

Resource nationalism and the value of natural resources to the countries in which they are located increase the likelihood of government interference or unfavorable changes in taxes, royalties, mining regulations, or agreements.

Other considerations may include safety records and a mining company's focus on safety, which also can boost productivity. The mining industry presents a number of operating risks, particularly in underground mines but also in surface mines and processing facilities. Companies with the financial capacity to invest in and use more advanced technology, such as autonomous haul trucks and haulage systems, are likely to see improvement in safety records and productivity.

Liquidity

Liquidity is an important rating consideration for all mining companies, and extremely weak liquidity can heavily affect ratings in many cases. However, the relative strength or weakness of liquidity may not have a substantial impact in discriminating between two issuers with an otherwise similar credit profile. Liquidity can be particularly important where companies have large short-term demands on liquidity, for example in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely near-term liquidity requirements from the perspective of both sources and uses of cash. For more information on our approach, please see our liquidity cross-sector methodology.[5]

Regulatory considerations

Companies in the mining sector are subject to varying degrees of regulatory oversight. Effects of these regulations may entail limitations on operations, higher costs, and higher potential for technology disruptions and demand substitution. Regional differences in regulation, implementation or enforcement may advantage or disadvantage particular issuers.

Our view of future regulations plays an important role in our expectations of future financial metrics, as well as our confidence level in the ability of an issuer to generate sufficient cash flows relative to its debt burden over the medium and longer term. Regulatory considerations also play a role in our assessment of an issuer's business profile. In some circumstances, regulatory considerations may also be a rating factor outside of the scorecard, for instance when regulatory change is swift.

Reserves

The level of proven and probable reserves is an estimate of the amount of ore that can be economically and legally extracted. We generally focus on proven and probable reserves rather than possible or inferred reserves given the greater uncertainty around possible or inferred reserves. It therefore gives a sense of the likely longevity of the production profile, which can be an important credit consideration, especially for speculative grade companies with few operating mines. Key characteristics affecting reserve quality include ore grade and recovery rates, metal price assumptions, geological data obtained and whether the reserves are associated with existing mining operations or require greenfield development. The underlying geology of a deposit is also important because it significantly affects mineral recovery rates and grades as well as development and operating costs. The location of reserves, including whether the deposit is a surface or underground mine, as well as altitude and proximity to existing infrastructure and end-markets may also be considered.

However, given the number of assumptions involved, the varying time requirements for permitting and development and the potential for changing permit requirements, the level of reserves is not necessarily a good predictor of future cash flow. Also, the levels of proven and probable reserves, grades, recovery rates and reserve life vary greatly among base metals, precious metals, iron ore, coal and other minerals. In addition, various reserve-testing requirements apply to the industry depending on the country of domicile. Consequently, we typically consider reserve quality within the context of a specific mineral, the company's reporting requirements and peer comparisons.

Although the scorecard does not have a factor that directly assesses proven and probable reserves, a company's ability to develop reserves into mineral production and to maintain or grow production levels ultimately affects its earnings performance and cash flow generation and can therefore have an impact on scorecard metrics such as EBIT margin and RCF/Debt. Reserve levels in themselves do not generally differentiate credit profiles, but they can be a more important consideration for companies with few operating mines or if reserve life is limited for the key producing mines. In these cases, weakness in reserve levels may be considered outside of the scorecard and can potentially limit a rating's upward movement.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies and in evaluating management performance relative to the performance of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that are far in excess of their operating needs. This excess cash, together with the company's policy and motivations, can be an important credit consideration.

The level of cash required to run a business can vary materially based on the region(s) of operation and the specific sub-sectors in which the issuer operates. As a result, cash on the balance sheet is most often considered qualitatively, by assessing the company's track record and financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We analyze possible uses of cash, especially for companies that may be under shareholder pressure to return it. By contrast, some companies maintain large cash holdings for long periods in excess of their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some beneficial cushion against credit deterioration, and cash balances are often considered in our analysis of near-term liquidity sources and uses. For more information, please see the "Liquidity" section.

While the scorecard in this methodology uses leverage and coverage ratios with total (or gross) debt, in cases where we are sufficiently confident that cash will remain on the balance sheet or will be used for creditor-friendly purposes, we may consider excess cash in a more quantitative manner. We do not typically adjust the balance-sheet debt for any specific amount because this implies greater precision than we think is achievable for the uncertain future uses of cash. We may consider creditor-friendly use of cash in our forward view of metrics, for instance, when we expect the cash to be used for debt repayment. We may also cite rating threshold levels for certain companies based on net debt ratios, particularly when these issuers have publicly stated financial targets based on net debt metrics.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this industry; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important to our forward view of metrics that are in the scorecard or other rating factors.

For example, free cash flow is not always an important differentiator of credit profiles. Strong companies with excellent investment opportunities may demonstrate multiyear periods of negative free cash flow while retaining solid access to capital and credit, because these investments will yield stable cash flows in future years. Weaker companies with limited access to credit may have positive free cash flow for a period of time because they have curtailed the investments necessary to maintain their assets and future cash-generating prospects. However, in some cases, free cash flow can be an important driver of the future liquidity profile of an issuer, which, as noted above, can have a meaningful impact on ratings.

Non-wholly owned subsidiaries

Some companies in the mining sector choose to dilute their equity stake in certain material subsidiaries, for example through an initial public offering, which may in some cases negatively impact future financial flexibility. While improving cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative impact on cash flows available to the parent company that may not be fully reflected in consolidated financial statements. For example, in the case of an equity stake reduction in a subsidiary down to 75%, in the parent's financial statements, all revenue and EBITDA of the subsidiary would typically still be consolidated at the group level.

The parent's share of dividend flows from a non-wholly owned subsidiary is reduced, and minority stakes can increase structural subordination, since dividend flows to minority interest holders are made before the cash flows are available to service debt at the parent company. While less frequent, sale of a minority stake may be accompanied by policies protective of the subsidiary that further limit the parent's financial flexibility, for instance, restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. However, in some cases we may find that an additional view of financial results, such as analyzing cash flows on a proportional consolidation basis, may be very useful to augment our analysis based on consolidated financial statements. When equity dilution or structural subordination arising from non-wholly owned subsidiaries is material and negative, the credit impact is captured in ratings but may not be fully reflected in scorecard-indicated outcomes.

For companies that hold material minority interest stakes, consolidated funds from operations typically includes the dividends received from the minority subsidiary, while none of its debt is consolidated. When such dividends are material to the company's cash flows, these cash flows may be subject to interruption if they are required for the minority subsidiary's debt service, capital expenditures or other cash needs. When minority interest dividends are material, we may also find that proportional consolidation or another additional view of financial results is useful to augment our analysis of consolidated financials. We would generally also consider structural subordination in these cases. Proportional consolidation brings a portion of the minority subsidiary's debt onto the balance sheet, but this debt is structurally senior to debt at the parent company, because it is closer to the assets and cash flows of the minority subsidiary. When these credit considerations are material, their impact is captured in ratings but may not be fully reflected in scorecard-indicated outcomes.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in an issuer's fundamental creditworthiness, which may cause actual ratings to be lower than the scorecard-indicated outcome. Event risks — which are varied and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized firm. Some other types of event risks include M&A, asset sales, spinoffs, shareholder distributions, litigation, pandemics, significant cybercrime events and geopolitical conflicts.

Parental support

Ownership can provide ratings lift for a particular company in the mining sector if it is owned by a highly rated owner(s) and is viewed to be of strategic importance to those owners. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic incentives to provide support to the issuer in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or has already done so in the past. Conversely, if the parent puts a high dividend burden on the issuer, which in turn reduces its flexibility, the ratings would reflect this risk.

Government-related issuers may receive ratings uplift due to expected government support. However, for certain issuers, government ownership can have a negative impact on the underlying Baseline Credit Assessment.[6] For example, price controls, onerous taxation and high distributions can have a negative effect on an issuer's underlying credit profile.

Other institutional support

In some countries, large corporate issuers have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. In Japan, our corporate ratings consider the support that has operated there for large and systemically important organizations. Over the years, this has resulted in lower levels of default than might otherwise have occurred. Our approach considers whether the presence of group and banking relationships may provide support when systemically important companies encounter significant financial stress.

Using the scorecard to arrive at a scorecard-indicated outcome

1. Measurement or estimation of factors in the scorecard

In the "Discussion of the scorecard factors" section, we explain our analytical approach for scoring each scorecard factor or sub-factor, and we describe why they are meaningful as credit indicators. When a factor comprises sub-factors, we score at the sub-factor level. Some factors do not have sub-factors, in which case we score at the factor level.

The information used in assessing the sub-factors is generally found in or calculated from information in the company's financial statements or regulatory filings, derived from other observations or estimated by Moody's analysts. We may also incorporate non-public information.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends of a company's performance, as well as for peer comparisons. Financial metrics,[7] unless otherwise indicated, are typically calculated based on an annual or other 12-month period. However, the factors in the scorecard can be assessed using various time periods. For example, rating committees may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics incorporate our standard adjustments[8] to income statement, cash flow statement and balance sheet amounts for items such as underfunded pension obligations and operating leases. We may also make other analytical adjustments that are specific to a particular company.

2. Mapping scorecard factors to a numeric score

After estimating or calculating each factor or sub-factor, each outcome is mapped to a broad Moody's rating category (Aaa, Aa, A, Baa, Ba, B, Caa, Ca or C, also called alpha categories) and to a numeric score.

Qualitative factors are scored based on the description by broad rating category in the scorecard. The numeric value of each alpha score is based on the scale below.

Exhibit 3
Numeric equivalents for qualitative factor and sub-factor scores

Aaa	Aa	A	Baa	Ba	B	Caa	Ca	C
1	3	6	9	12	15	18	20	21

Source: Moody's Ratings

Quantitative factors are scored on a linear continuum. For each metric, the scorecard shows the range by alpha category. We use the scale below and linear interpolation to convert the metric, based on its placement within the scorecard range, to a numeric score, which may be a fraction. As a purely theoretical example, if there were a ratio of revenue to interest for which the Baa range was 50x to 100x, then the numeric score for an issuer with revenue/interest of 99x, relatively strong within this range, would score closer to 7.5, and an issuer with revenue/interest of 51x, relatively weak within this range, would score closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the value of the metric that constitutes the lowest possible numeric score, and the value that constitutes the highest possible numeric score).

Exhibit 4
Ranges of numeric equivalents for quantitative factor and sub-factor scores

Aaa	Aa	A	Baa	Ba	B	Caa	Ca	C
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5	20.5 - 21.5

Source: Moody's Ratings

3. Determining the overall scorecard-indicated outcome

The numeric score for each sub-factor (or each factor, when the factor has no sub-factors) is multiplied by the weight for that sub-factor (or factor), with the results then summed to produce an aggregate numeric score. The aggregate numeric score is then mapped back to a scorecard-indicated outcome based on the ranges in the table below.

Exhibit 5
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

For example, an issuer with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the corporate family rating (CFR) for speculative-grade issuers and to the senior unsecured rating for investment-grade issuers. For issuers that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or Baseline Credit Assessment for comparison to the scorecard-indicated outcome. For an explanation of the Baseline Credit Assessment, please refer to *Rating Symbols and Definitions* and to our cross-sector methodology for government-related issuers.[9]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a corporate family rating (CFR) to speculative-grade issuers or a senior unsecured rating for investment-grade issuers. For issuers that benefit from rating uplift from government ownership, we may assign a Baseline Credit Assessment.[10]

Individual debt instrument ratings may be notched up or down from the CFR or the senior unsecured rating to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. The documents that provide broad guidance for such notching decisions are the rating methodology on loss given default for speculative-grade nonfinancial companies, the methodology for notching corporate instrument ratings based on differences in security and priority of claim, and the methodology for assigning short-term ratings.[11]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, please see *Rating Symbols and Definitions*.[12]

Limitations

In the preceding sections, we have discussed the scorecard factors and many of the other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall rating methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to actual ratings.

The scorecard in this rating methodology is a relatively simple tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings to these companies. Credit loss and recovery considerations, which are typically more important as an issuer gets closer to default, may not be fully captured in the scorecard. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for issuers at the upper and lower ends of the rating scale.

The weights for each factor and sub-factor in the scorecard represent an approximation of their importance for rating decisions across the sector, but the actual importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors that are outside of the scorecard, including those discussed above in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[13] Examples of such considerations include the following: how sovereign credit quality affects non-sovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking time periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology document does not include an exhaustive description of all factors that we may consider in assigning ratings in this sector. Companies in the sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for an issuer's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

Moody's related publications

Credit ratings are primarily determined through the application of sector credit rating methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

For data summarizing the historical robustness and predictive power of credit ratings, please click here.

For further information, please refer to *Rating Symbols and Definitions*, which is available here.

Endnotes

1 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 For definitions of our most common metrics, please see our cross-sector methodology that describes our financial statement adjustments in the analysis of nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 For definitions of our most common metrics, please see our cross-sector methodology that describes our financial statement adjustments in the analysis of nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

4 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

5 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

6 For an explanation of the Baseline Credit Assessment, please refer to *Rating Symbols and Definitions* and to our cross-sector methodology for government-related issuers. A link to *Rating Symbols and Definitions* and a link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

7 For definitions of our most common metrics, please see our cross-sector methodology that describes our financial statement adjustments in the analysis of nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

8 For an explanation of our standard adjustments, please see our cross-sector methodology that describes our financial statement adjustments in the analysis of nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

9 A link to *Rating Symbols and Definitions* and a link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

10 For an explanation of the Baseline Credit Assessment, please refer to *Rating Symbols and Definitions* and to our cross-sector methodology for government-related issuers. A link to *Rating Symbols and Definitions* and a link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

11 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

12 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

13 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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Authors

Jamie Koutsoukis
Dirk Steinicke